INDEPENDENT AGENT REVIEW OF THE DEGREE-10 AND DEGREE-11 CLASSIFICATIONS
28 October 2026

Review status: COMPLETE. No blocking mathematical or certificate gap found.
The exact final evidence scope is final_evidence_manifest.json.

The reviewed claims are

$$S(10) = \{2,3,5,7,11,13,17,19,23,29,31,37\},$$
$$S(11) = \{2,3,5,7,11,13,17,19,23,29,31\},$$

for arbitrary elliptic curves over number fields of EXACT degree 10 and 11, respectively. No assumption that the curve is defined over Q is made.

The original proof/archive versions are identified by SHA256 in review_scope_snapshot.json. The reviewed documents were outputs/S10_proof_audited.txt and outputs/S11_proof_audited.txt, with their final work/audit10 and work/audit11 sources/data and final archive manifests. Earlier superseded nonic/decic proofs and the case-ledger PASS summaries were not used as mathematical hypotheses or substitutes for checking the calculations.

Separate verdicts

S(10): The claimed classification withstands this independent review as a computer-assisted proof on the stated external-theorem and software dependencies. No unresolved prime, reduction type, or certificate failure was found. The degree-10 occurrence at 37 is valid, so Primes(31) would be an incorrect degree-10 classification. The final Primes(37) claim is supported.

S(11): The claimed classification withstands this independent review as a computer-assisted proof on the stated external-theorem and software dependencies. All 73 selected large-character tests were independently rerun, and their scalar complexities were independently certified outside Magma’s recurrence routine. No unresolved prime, reduction type, or certificate failure was found.

Two small exposition repairs and one recommended explanatory expansion are specified below. They do not require a changed computational certificate or an additional mathematical exclusion case.

This was an independent AI-agent review, with two focused child reviewers. It is NOT external human peer review and NOT formal verification in a proof assistant. The original proof and certificate files were left untouched.

Evidence actually regenerated

1. Occurrence and exact degrees.
All seven degree-11 witnesses for 11,13,17,19,23,29,31 and the degree-10 witnesses for 23,29,31,37 were rerun exactly. Fields have the claimed degree; the curves are nonsingular; P is nonzero and pP=O. Exact order follows since p is prime. The degree-10 37 witness has j-degree 10, and is valid for the arbitrary-curve problem. Occurrence through 19 in degree 10 accepts the known quintic classification and quadratic base extension.

For a K-valued modular point the closed residue degree divides [K:Q]. For degree 11, Mazur excludes residue degree1 for p>7. For degree 10, the known S(1),S(2),S(5) bounds exclude proper divisor degrees for p>19. This justifies the exact-degree trace divisors and the S_new(d) hypotheses in Derickx--Stoll; it does not equate degree 10 and degree 11 points.

2. Global bound and character tests.
The primary Derickx--Stoll statements were read and checked for the precise small-prime, CM, divisibility, gonality and rational-2-primary-torsion hypotheses used. Table2 was compared with the actual primary arXiv-v1 HTML, and the arithmetic exclusions were independently recomputed: 5987 primes in the degree 10 table range and 9560 in the degree 11 range. The table is not extrapolated beyond 100000 and no asymptotic conjecture is assumed.

The 52 rank 11 matrices over F3 with M53 and 72 over F5 with M73 were independently constructed and verified. The primary Parent criterion really does allow those odd characteristics. The Hasse/toric sieve above 100000 was independently reconstructed for all 6567 primes up to 177989: 6522 are excluded, 45 remain, and the fresh characteristic 3 j=0 twist enumeration removes 24. The remaining21 primes and exact selected order coverage give exactly 73 large-character tests.

The mathematical character argument was checked in addition to arithmetic. The Eisenstein factors remove the boundary; the cuspidal winding module has full dimension if an exact scalar Krylov sequence has that linear complexity; reduction gives a rank lower bound in characteristic zero; and Galois conjugacy covers every character of a fixed order. The selected missing r-adic power in str(p), with r!=11, excludes the divisibility alternative. The genuinely nonreal boundary control p176321/order17632 has chi(2) of order 551 and directly verifies that the full boundary is killed and the modified winding vector is cuspidal. A separate small-level control distinguishes the stated chi convention from its inverse.

All 73 native character tests were freshly rerun with seed 20261029 instead of 20261028; each has full scalar complexity equal to its cuspidal dimension. The scalar Krylov sequences were exported, and all 73 complexities were independently certified in Sage by polynomial Euclid and exact coprime Pade certificates, rather than reusing Magma’s Berlekamp--Massey routine. The retained numerator/denominator polynomials and sequence hashes permit direct checking of the lower bound. The complete native batch took about 713 seconds; the independent sequence checker overlapped it. Final outputs include global_fi/fresh_73_results.json, global_fi/independent_sequence_results.json, global_fi/replay_73.log, global_fi/check_sequences.log, and the 73 native sequence files and 73 independent Pade certificate files under global_fi/large73.

The remaining small global primes are covered by the stated diamond-orbit and gonality argument; its finite-image and 2-primary torsion hypotheses are supplied by the cited unconditional results. The final caps are p<=101 for degree 10 and p<=103 for degree 11.

3. Rational cusps and formal immersion.
All 16 actual DKSS endomorphism exports were regenerated. Every raw operator matrix and every injective row compression matched the original. A freshly compiled checker reran all 2717157222 normalized cusp multiplicity tuples, including 2580289822 at83. Independent bounded-composition inclusion-exclusion counts agree in every case. The combined-map projections preserve the required implication from projected to unprojected independence.

The endomorphisms annihilate the rational Jacobian groups, not merely their free parts: the winding test gives finite image and the odd Hecke factor kills rational torsion. Matrix independence on an integral modular-symbol lattice implies the needed Hecke-algebra independence modulo2. This invokes the DKSS endomorphism criterion, without assuming that the cotangent lattice of an optimal winding quotient has an unproved property in characteristic 2.

In degree 10, augmenting a maximal multiplicity by1 embeds each degree 10 tuple into a successful degree 11 tuple; its ten operators form an independent subset. The annihilators are independent of total degree. This transfer is valid and does not assume the S(11) conclusion.

4. Complete finite-field support census.
A separately written census used explicit ordinary twists, EVERY j in every field F_(2^d), 1<=d<=11, all supersingular twists, and a different Frobenius key. All 176 (level,degree) pairs agree, not just aggregate totals: 210 diamond orbits, of which206 are ordinary and 4 supersingular, and 6092 closed points. Proper j-fields are retained. The p^2 nondivisibility checks make the rational p-torsion cyclic, so all nonzero scalars of one generator cover all relevant level structures. The fine-moduli property and complete twist enumeration account for supersingular automorphisms and descent.

At 41 the ordinary minimum is8, while the degree 5 points are supersingular. Hence the only possible multiple-noncuspidal support in total degree 10 or11 is the degree 5+5 case at 41, including repetition. The nonrational cusp degrees are too large to coexist with a noncuspidal point; all other complements are rational cuspidal. This checks the actual coverage, not only census markers.

5. Full integral models and ordinary Hecke obstructions.
All 16 canonical q-bases, unit relations, ordinary geometric3-isogeny targets, trace weights and eight-term jets were freshly regenerated from the source. Changed bases at 41/73/97 were independently matched by Sage: the full q-row spaces agree, diamond matrices are conjugate, and unit-relation spaces are equivalent. The other raw outputs match byte-for-byte.

The full integral differential-lattice assertion was checked mathematically. Fricke sends (E(q^p),q) to E(q) with its dual mu_p level structure, identifies the formal parameter q, and extends over Z2 because p is odd. Thus the rational q-form space is the actual differential space on the point-level curve. Every stored precisionB satisfies B-1>2g-2. If a rational2-adic differential has an integral B-term truncation but is not integral, multiplying by its minimal denominator gives a nonzero mod2 holomorphic differential vanishing to order at least B-1>2g-2, impossible. Therefore saturation of those truncations is the full Z2 lattice. This is a justification of the model, not an inference from successful finite-field ranks.

The modular units’ lack of interior poles was independently checked from the exact low-order division-polynomial factors (orders2,3,4,5,7,8,9). Every rational and nonrational cusp was used in the pole calculations. The global zero-order bound then proves each unit relation identically: a nonzero meromorphic differential cannot have more zeros than 2g-2 plus its total pole degree. At every ordinary source/target, rankg-1 of the combined relations gives the actual canonical point. The verified dimension 3g-3 of canonical products also rules out hyperellipticity, so canonical keys really separate points.

The inverse-diamond convention and geometric A3=T3-1-3delta3 were checked. On rational Tate cusps, one cyclic3-subgroup gives the same cusp and three give delta3 of the cusp, so A3 kills the actual divisor. Eichler--Shimura and odd-prime torsion reduction give its rational-torsion annihilation. If a source has degreed, a3-kernel factor has degreee, and the target moduli field has degreet, its trace weight is de/t; all factors, repeated targets and cancellations are retained. The total Hecke degree is4d.

Alpha finiteness uses the actual rankzero list and strange character orders:

sigma^6-1 at 61, (sigma-1)Phi12(sigma) at 97 and (sigma-1)(sigma^5+1) at 101 kill all potentially positive-rank character factors; Kato gives finite image on the remainder. This has no dependence on total degree 10 versus11.

The non-principality logic is correct. If alpha*A3*P=U-V=div(g) and a nonzero F in R=H^0(K^k) vanishes onU, then S=F/g is a nonzero section vanishing onV. For every H inR vanishing onU, H*S lies in F*R. The recorded homogeneous linear system gives necessary conditions onS; full rank contradicts its existence. Random H can fail to certify an obstruction but cannot create a false one. Multiplication tensor identities are checked beyond the appropriate Sturm bounds, and jets are obtained by checked Hensel lifting in the actual ordinary parameter a6=1/j.

With a changed seed all 202 ordinary obstructions were freshly regenerated. Sage independently checked all 202 ORIGINAL and all 202 FRESH matrix ranks. The same four level 37 exceptions remain, and are separately excluded below. The 96 ordinary orbits required in degree 10 are a subset of this same actual through-degree 11 census; alpha*A3*P is independent of total degree.

6. Exceptional branches and gonality43.
All 19 assigned native computations were freshly rerun, including the 41/43 supersingular obstructions, all 41 supersingular pairs, the 41 nonrational cusp class, the 37 stabilizers and exceptional degree 10 witness match, the actual JOne integral2-torsion calculations at 37/43/73, all 18 exceptional37 cotangent restrictions, and all 8 nonrational-pole Riemann--Roch cases at43.

The integral homology really belongs to JOne rather than an unspecified isogenous Jacobian; the relevant Magma implementation was inspected. Fresh common-kernel dimensions are0/6/6 at 37/43/73 and the required diamonds act trivially. The37 cotangent image has dimension 38 and all 18 restrictions to the decic divisor plus a rational cusp have rank 11. The zero rational 2-torsion and formal-immersion uniqueness argument therefore excludes that last irreducible degree 11 branch; it does not wrongly exclude the actual degree 10 witness.

The stabilizer/norm argument is valid, including the 2-primary reduction kernel and each gonality inequality. A degree-zero interpretation of its printed shorthand is specified below. The89 branch correctly yields a rational X0(89) point and invokes the prime-isogeny theorem. The published level 37 gonality 18 is the correct external input.

The new level 43 gonality proof was independently reconstructed from the raw canonical q-basis, using Sage right kernels and reversed cusp indexing. All234 envelopes cover all 66 pole-multiplicity partitions; the residue pairing is precisely the global principal-parts obstruction. Of the envelopes, 229 have zero kernel, four have dimension 1 and one has dimension 4. Every nonzero principal part has pole degree at least 30, excluding degree<=15. The native first-envelope control gives the same 15 pole degrees. Native nonrational-pole cases all have Riemann--Roch dimension 1, and the degree 8 points form the two required length21 diamond orbits. Thus gon_F2>=16, and good-reduction specialization gives gon_Q>=16. This also supplies the bound needed by the degree 10 transfer, without trusting a stronger website claim.

Precise proof-text clarifications

A. outputs/S11_proof_audited.txt:224-228. Add: ’At 41 the degree 5 points are supersingular and the minimum ordinary degree is8.’ The currently printed minimum 5 statement alone would not exclude a hypothetical5+6 support in degree 11; the actual complete census excludes it. The degree 10 sentence at

outputs/S10_proof_audited.txt:113-116 remains valid but can repeat this fact.

B. outputs/S11_proof_audited.txt:337-344, especially line340, and the corresponding degree 10 stabilizer passages. Write

$$y=[(h-1)(D-C)] \text{ in } J(Q),$$

rather than treating [D-C] itself as a degree-zero Jacobian point. For the torsion deduction put Z=D-C and N_hZ=sum_(i=0)^(m-1) h^iZ. Then mZ-N_hZ has degree zero and

$$m\,y=(h-1)[mZ-N_hZ].$$

Finite (h-1)J(Q) makes my, hence y, torsion. The norm telescopes exactly and the subsequent reduction/norm argument is unchanged. This supplies the missing degree bookkeeping without an additional hypothesis.

C. outputs/S11_proof_audited.txt:278-286. Recommend expanding the saturation claim with the B-1>2g-2 minimal-denominator argument above. The current q-expansion/Fricke assertion is correct on standard geometric facts; the explicit lemma makes its full integral-lattice premise reviewable and prevents confusing saturation of an arbitrary finite truncation with proof of integrality. No change to raw models or ranks is needed.

Dependencies and limits of the verdict

The accepted mathematical inputs are Mazur’s rational torsion and prime-isogeny theorems; the known quintic and quadratic prime classifications; DKSS’s Oesterle bound, rational-torsion annihilators, rankzero list and endomorphism formal-immersion criterion; Parent’s odd-characteristic criterion and2-primary reduction-kernel result; the unconditional Derickx--Stoll theorems and finite strange-character Table2; Kato’s rankzero theorem; the published X1(37) gonality; and standard smooth-model, q-expansion, Fricke, Eichler--Shimura, principal-parts duality and gonality-specialization facts. The primary statements and hypotheses used were checked. The entire published strange-character table below100000 was NOT independently recomputed; it is an explicit external mathematical input.

The software trust includes Magma V2.30-1 exact arithmetic and its modular symbols, q-eigenforms, finite-field twists/groups, isogenies, Jacobians and function-field/Riemann--Roch routines; the bundled modular-curve library; Sage’s exact finite-field linear algebra/polynomials; and the reviewed Python/C++ orchestration. Independent implementations and changed seeds substantially check the certificates but do not reimplement those CASs or formally prove them correct. A successful replay is not a substitute for the mathematical model/coverage arguments given above.

Final evidence and reproduction

The final_evidence_manifest.json identifies exact final sources, logs and data with SHA256 and byte counts; files not listed there are not final evidence. The focused reports are ordinary/REVIEW.txt, global_fi/REVIEW.txt and exceptional/review.txt. They give source/primary-theorem line references and the detailed mathematical checks. Original computational inputs are supplied by the unchanged audited archives recorded in review_scope_snapshot.json.

Reproduction requires that original source tree, Magma V2.30-1, Sage’s Python, and a C++17 compiler. Review drivers retain workspace/library paths, so those paths must be relocated when replaying on another computer. The final manifest includes the exact commands and dependency hashes, and a verification that the original proof/archive snapshot was unchanged at review completion.

The proposed revisions in proposed/S10_proof_audited.txt and

proposed/S11_proof_audited.txt, with their unified diffs, were also read.
They correctly implement A--C above, including ’after tensoring with Z_2’
in the full-lattice conclusion. They introduce no changed computational
hypothesis or certificate. I approve those specific mathematical clarifications;
their exact reviewed hashes are retained in the final evidence manifest.